

Md. Sohanur Rahman

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SUMMARY

With nearly 7 years of professional experience in the IT industry, I specialize in backend engineering, AI-native systems, and full-stack development — building modern, large-scale applications that span intelligent automation to cloud-native infrastructure.

- Proficient in building scalable backend solutions with FastAPI, Django & Laravel with a strong focus on RESTful API design, domain-driven design (DDD), and test-driven development (TDD) — delivering clean, maintainable codebases across diverse PHP-driven projects
- Deep expertise in AI-native and agentic systems, including Graph RAG architectures, Model Context Protocol (MCP), LangChain, and LangGraph for building context-aware, graph-driven intelligent applications.
- Hands-on experience developing and deploying machine learning models and data-driven solutions, covering EDA, dataset preparation, model training, and semantic search with vector databases (Pinecone, FastEmbed, HuggingFace)
- Proficient in building scalable backend solutions with FastAPI, Django, and Laravel, with a strong focus on RESTful API design, domain-driven design (DDD), and test-driven development (TDD)
- Experienced integrating Large Language Models (LLMs) into production systems for intelligent automation, conversational querying, and context-aware responses
- Skilled in full-stack development — crafting responsive frontend interfaces using React and TypeScript alongside robust API backends
- Proficient in microservices architecture and containerized deployments using Docker and Kubernetes on cloud platforms (Azure)
- Expertise in both structured and unstructured databases: PostgreSQL, SQLite, MongoDB, and graph databases (Neo4j with Cypher Query Language)
- Experienced with CI/CD pipelines, Nginx, and automated deployment processes following Agile and DevOps best practices
- Strong background in scripting and command-line tooling (shell, Python scripting) for automation and data pipeline workflows
- Collaborates effectively in cross-functional teams and actively mentors junior developers
- Strong problem-solving instincts with a consistent record of delivering high-quality software in complex domains

SKILLS

Backend: Python, Django, FastAPI, Celery, Redis, PHP, Laravel, WebSocket, OpenCV, Selenium, LangChain, LangGraph, SQLAlchemy, spaCy, PyTorch

AI & Machine Learning: LLM Integration, Graph RAG, Model Context Protocol (MCP), Vector Search & Embeddings, Semantic Search, Pinecone, FastEmbed, HuggingFace, Neo4j (Cypher), Exploratory Data Analysis (EDA)

DevOps: Docker, Kubernetes, AWS (EC2, S3, API Gateway), Azure, CI/CD, Nginx, Amazon S3

Frontend: JavaScript, React, TypeScript, jQuery, HTML, CSS, Web3

Databases: PostgreSQL, SQLite, MongoDB, Neo4j, Redis

Tools & Practices: Git, Shell Scripting, REST API Design, Microservices, TDD, DDD, Agile

PROFESSIONAL EXPERIENCE

Senior Software Engineer II | **Cefalo** | Dec 2024 – Present

Technology: Python, FastAPI, Neo4j, LangChain, LangGraph, Graph RAG, MCP, React, TypeScript, Docker, Vector Search & Embeddings, Cypher Query Language, PyTorch, spaCy

- Promoted to Senior Software Engineer II, contributing to AI-native product development within cross-functional teams
- Developed an internal employee graph application enabling complex, conversational queries using Graph RAG and Model Context Protocol (MCP)
- Designed a FastAPI backend with Neo4j for graph storage, supporting intelligent query routing, auto-generated Cypher queries, and multi-provider LLM integration
- Implemented vector search and semantic embeddings (FastEmbed, HuggingFace) for context-aware employee information retrieval and summarization

Senior Software Engineer I | **Cefalo — Wirescan AS, Norway (Remote)** | *Dec 2024 – Present*

Technology: *Python, PostgreSQL, Redis, Mapbox, Docker, CI/CD, AG-Grid*

- Collaborated on digitising and optimising power cable maintenance and data onboarding workflows for offshore and energy clients
- Designed and implemented an onboarding portal to automate extraction of cable and turbine data from diverse file formats (PDF, CSV, Excel), reducing manual input efforts by 90%
- Developed real-time data monitoring dashboards using Dash and Plotly, enabling dynamic visualization of key performance metrics for field operators and analysts
- Built interactive geographic visualizations using Mapbox for cable infrastructure, improving spatial awareness of assets like turbines, substations, and joints
- Implemented high-performance AG-Grid data tables, signal analysis tools, and event inspection features to support diagnostic workflows
- Created journal matrix systems, exportable reporting modules, and step-by-step model builders for streamlined cable onboarding and assessment

Software Engineer | **SELISE Digital Platforms** | *April 2022 – May 2024*

Technology: *Docker, Python, Django, Celery, Redis, FastAPI, Microservice, NodeJS, ExpressJS*

- Worked directly with Swiss clients on business requirement analysis, design, and software development
- Wrote and tested code, refined and rewrote as necessary to meet quality standards
- Designed and developed integrated Django, Shopify, and Wagtail CMS solutions using proper conventions
- Led the development team and worked closely with cross-functional staff
- Maintained systems by monitoring and correcting software defects; handled server deployments

Backend Engineer (Part-Time) | **Meghna Cloud** | *June 2023 – Dec 2024*

Technology: *Cloud, Django, Python, OpenStack, Channels, Redis, Microservice, Kubernetes, Apache Airflow*

- Analyzed new requirements and explored potential solutions, including optimal architectural design
- Participated actively in coding, evaluated team contributions, and mentored junior developers
- Developed and maintained APIs to support application functionality
- Handled server deployment and implemented CI/CD processes for continuous integration
- Worked with the NOC and SOC teams to identify and implement effective solutions
- Managed data pipelines and workflow automation using Airflow to ensure data integrity and smooth data processing

Lead Software Engineer | **Cramstack** | *September 2021 – April 2022*

Technology: *Docker, Amazon Web Services (AWS), Web3, Django, Amazon S3, AWS API Gateway*

- Designed and developed SaaS-based decentralised Web3 applications
- Wrote testable, scalable, and efficient code

Senior Python Developer | **Bdtask Ltd.** | *February 2021 – September 2021*

Technology: *Python, JavaScript, jQuery, Linux, Rest API, AWS Lambda*

- Worked closely with stakeholders to understand business requirements
- Application design and development, bug fixing
- Used user stories and agile methodologies to evaluate problem scenarios and design feasible solutions

Jr. Software Engineer | **Codesmith Tech Ltd.** | *September 2019 – February 2021*

Technology: PHP, Laravel, *Python*, *JavaScript*, *jQuery*, *Django*

- Project planning, database design, and developing software solutions based on client requirements
- Practiced proper coding conventions and agile methodologies
- Analyzed, diagnosed, and resolved errors related to applications

EDUCATION

Master's Degree — University of Rajshahi | 2015 – 2017 | *Thesis: Parallel Programming with Clustering*

Bachelor's Degree — University of Rajshahi | 2011 – 2015

CERTIFICATION & ACHIEVEMENT

Participated in a 48 hours **AI Hack-fest** at Cefalo Bangladesh Ltd.

OWASP Top 10:2021 in Python — Secureflag (Completed: Mar 2025)

Laravel & PHP Mastery: Build 5 Real-World Projects - (Completed: April 2025)

ITEE FE (Half Passer, March 2019): National Level IT Engineers Examination governed by IPA, Japan

Arduino and Robotics Examination (Rank: 2): Bangladesh Hi-tech Park Authority, Data Soft MultiTech, Rajshahi